

Non-Intrusive Chiller Load Monitoring Using XGBoost and Meteorological Context

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Abstract—Physical sub-meters in building energy management systems can fail silently, producing weeks of false data with no automated alert. This paper presents a hybrid Non-Intrusive Load Monitoring approach for monitoring a centrifugal chiller in a university building using only the main building meter and publicly available weather data. Working with 239,447 historic energy consumption records from the Faculty of Building Services building of the Technical University of Cluj-Napoca between 2017 and 2026, the present paper proposes a multi-stage pipeline: three-step contextual data cleaning, Isolation Forest anomaly detection, Pruned Exact Linear Time change-point segmentation, Symbolic Aggregate approXimation symbolic analysis, and an XGBoost virtual sensor with 24 engineered features, including lag variables and Heat Index. The virtual sensor achieves $R^2=0.939$, $MAE=0.040$ kW, $F1=0.937$, and $AUC=0.999$ on the 2023–2025 test set – a 12-fold F1 improvement over the threshold baseline ($F1=0.075$). Results show that the virtual sensor’s distributed estimation error of 998 kWh over three years compares favorably to the 606 kWh silently misreported by the physical sensor during a single two-week freeze fault. The results of the new approach contribute to the improving the accuracy of equipment identification in university-scale BEMS.

Index Terms—Non Intrusive Load Monitoring, Energy Disaggregation, Anomaly Detection, Building Energy Management System, XGBoost

I. INTRODUCTION

Buildings consume roughly 40% of total energy in the EU, and reducing that figure depends largely on knowing where the energy goes [1]. Building Energy Management Systems (BEMS) address this challenge by monitoring and controlling individual sub-systems, but they rely on physical sensors – and physical sensors break [6].

In December 2017, the chiller sub-meter (ID_32) in the Faculty of Building Services building of the Technical University of Cluj-Napoca (UTCN) froze at a constant value for over two weeks. During that time, it reported 606 kWh of energy that was never actually consumed, equivalent to 163 kg of CO₂ in false accounting.

Non-Intrusive Load Monitoring (NILM) offers a practical alternative: instead of installing dedicated hardware for each load, the aggregate signal from a single main meter is analyzed to infer what individual devices are doing [3], [7]. The idea is straightforward, but applying it to a university chiller brings real challenges. The chiller is active only about 5% of the time, which makes every standard classifier struggle with class

imbalance. Classical thresholds achieve $F1=0.075$. That is not good enough.

A. Anomaly Detection in Energy Systems

Wang and Li [1] combine Canopy-Kmeans clustering with Isolation Forest [9] on Irish smart meter data to detect abnormal electricity users, achieving strong recall but without contextual validation using outdoor temperature, which is essential for chiller-specific cleaning.

Belgacem and Chihi [2] survey sensor fault detection across engineering domains and classify the frozen-sensor fault as a distinct fault type. Their taxonomy directly motivated the Stage 2 outlier removal methodology included in the present paper.

B. NILM and Load Disaggregation

De Diego-Otón et al. [3] apply supervised and unsupervised MobileNet architectures to event-based NILM on the WHITED dataset, achieving 95.76% accuracy on known appliances. Their work focuses on 5-second startup transients in laboratory conditions. The present study operates on 8 years of real operational data where the chiller runs for hours, not milliseconds.

Liu et al. [4] use XGBoost [8] with particle-swarm hyperparameter optimization to diagnose centrifugal chiller faults, reaching 99.67% accuracy on the ASHRAE RP-1043 dataset. This is the best fit to the present paper in terms of equipment, but their data is simulated in a controlled lab environment. The present study uses real campus data that includes sensor failures, missing periods, and operational anomalies.

C. Energy Forecasting in University Buildings

Nichiforov et al. [5] compared SARIMA and Recurrent Neural Networks for short-term load forecasting in seven university buildings at Politehnica University of Bucharest, showing RNN consistently outperforms SARIMA. Their work forecasts aggregate building consumption. In the present study, the proposed approach disaggregates individual equipment from that aggregate, which is a complementary and more advanced problem.

To the best of the authors’ knowledge, no existing work combines multi-stage contextual cleaning with a NILM virtual

sensor on real campus data over multiple years, with meteorological features (Heat Index, CDD) and a direct comparison against documented physical sensor faults.

The paper is organized as follows. Section II describes the dataset, preprocessing, and the full methodology. Section III reports and discusses results. The work ends with conclusions and future research directions.

II. PROPOSED METHOD

The following section presents the complete pipeline developed to address sensor inaccuracies in the BEMS infrastructure of the Faculty of Building Services building. The pipeline covers data cleaning, anomaly detection, seasonal analysis, and the construction of the XGBoost virtual sensor.

A. Dataset and Preprocessing

The dataset contains 239,447 records at 15-minute intervals from December 2017 to January 2026, from the Faculty of Building Services building of UTCN. Two channels are available: ID_31 (main building meter, mean 3.18 kW) and ID_32 (chiller sub-meter, mean 0.36 kW, operational range 2–15 kW when active). Outdoor temperature and relative humidity were downloaded from the Open-Meteo archive API and linearly interpolated to 15-minute resolution.

B. Why Standard Methods Fail on ID_32

The chiller is active for approximately 5% of all records. Applying IQR to ID_32 would flag normal summer operation (4–8 kW) as outliers because those values sit far above the overall distribution median. Isolation Forest has the same problem: with contamination set to match reality, it treats high summer values as anomalies. Therefore, the present paper proposes a domain-specific three-stage pipeline.

C. Three-Stage Contextual Cleaning

The first stage, Physical bounds, removes six records with $ID_{32} > 50$ kW as meter errors. The maximum observed value was 14,390 kW on 21 February 2022, which is physically impossible for this equipment.

The second stage, Frozen sensor detection, flags runs of eight or more consecutive identical values above 1.5 kW, outside summer months (June–August), as frozen-sensor events. This catches the December 2017 – January 2018 fault: 1,230 records where the meter reported a constant 2.027 kW for 12 consecutive days.

In the third stage, Contextual validation, values above 2.0 kW when outdoor temperature is below 15°C are removed. A cooling chiller has no physical reason to operate in cold weather. This removes 513 records distributed across autumn and winter months.

Table I summarizes the results. The 606 kWh falsely reported by the frozen sensor – equivalent to 163 kg CO₂ – would have gone undetected without this pipeline. Fig. 1 visualizes all three fault types.

TABLE I
OUTLIER REMOVAL SUMMARY – ID_32 CHILLER

Fault type	Records	% of total
Physical (>50 kW)	6	0.003%
Frozen sensor	1,230	0.514%
Contextual ($T < 15^{\circ}\text{C}$)	513	0.214%
Total	1,749	0.731%



Fig. 1. Three fault types in ID_32: physically impossible spikes (top, log scale), frozen sensor events (middle), and contextual outliers during cold weather (bottom).

D. Isolation Forest on the Main Meter

For ID_31, where class imbalance is not a problem, Isolation Forest (contamination=0.01) was applied and compared against IQR. Fig. 2 shows the result: IQR flags 4,915 records including 3,264 false positives (66% false positive rate), while Isolation Forest detects 2,394 real anomalies with zero false positives.

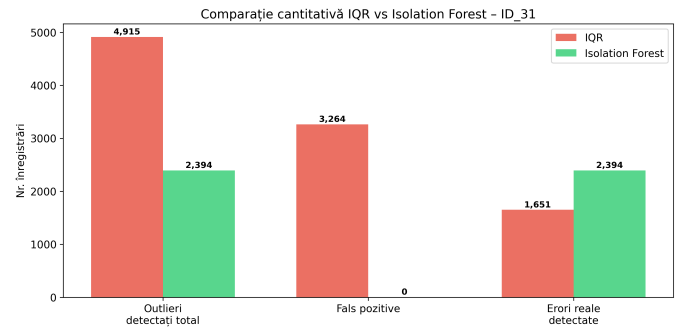


Fig. 2. IQR vs Isolation Forest on ID_31: IQR produces 3,264 false positives; Isolation Forest produces zero.

E. XGBoost Virtual Sensor

The virtual sensor predicts the instantaneous power of ID_32 from ID_31 and weather inputs. The ON/OFF state is derived by thresholding at 2.0 kW. Twenty-four features are divided into four groups:

The first group consists of ID_32 history (8 features): lag values at 15 min, 30 min, 1 h, 2 h, 12 h, 24 h; rolling means over 1 h and 4 h. These capture thermal inertia: if the chiller was running an hour ago, it is likely still running now.

The second group contains ID_31 history (6 features): lag values at 15 min, 1 h, 24 h; rolling means over 1 h and 4 h. The aggregate meter signal carries information about the building's overall thermal state.

The third group includes weather (5 features): outdoor temperature, temperature change over the last 15 min, 1-hour lagged temperature, relative humidity, and Heat Index.

The fourth group comprises calendar (5 features): sine and cosine encoding of hour and month – so that hour 23 is treated as adjacent to hour 0, not far from it – plus a binary working-day indicator.

The model is trained on 2018–2022 (131,656 records) and evaluated on 2023–2025 (101,984 records) with no data leakage between the two periods. XGBoost [8] parameters: $n_estimators=200$, $max_depth=8$, $learning_rate=0.05$, $subsampling=0.8$.

III. RESULTS AND DISCUSSION

A. Seasonal Profile

Fig. 3 shows the chiller's activity over the full eight-year period. It is active 5.4% of the time in summer, 0.6% in spring, 0.4% in autumn, and essentially never in winter. July peaks at 8.0%. This strong seasonality confirms that weather context is critical for any predictive model.

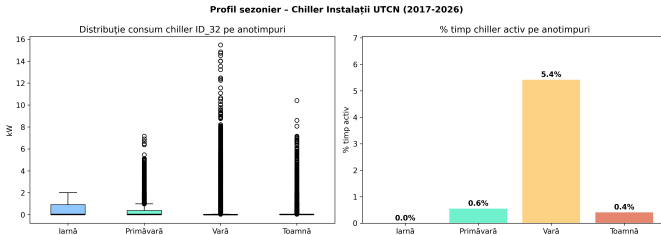


Fig. 3. Chiller seasonal profile (2017–2026): consumption distribution by season (left) and active time percentage by calendar month (right).

B. Monthly Consumption Heatmap

Fig. 4 presents the mean chiller power consumption per calendar month and year across the full 2017–2026 period. The heatmap confirms the strong seasonal concentration identified in the seasonal profile analysis. July 2022 stands out as the absolute peak of the entire series, consistent with the highest Cooling Degree Days recorded in that year. The years 2020–2021 appear as near-zero rows, corresponding to the frozen-sensor period identified during preprocessing. The year 2024

shows anomalously low consumption despite high CDD values, indicating an operational anomaly independent of climatic conditions. These observations validate the contextual cleaning decisions made in Section II and provide additional evidence for excluding 2020–2021 and 2024 from the correlation analysis.

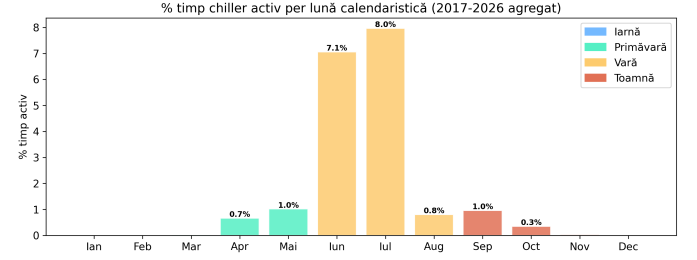


Fig. 4. Monthly mean chiller consumption heatmap (2017–2026). July 2022 represents the absolute peak. Years 2020–2021 show near-zero values due to the frozen-sensor fault.

C. Cooling Degree Days and Heat Index

To quantify the relationship between climate and chiller demand, CDD with base 18°C was computed in two variants: CDD_T using dry-bulb temperature only, and CDD_{HI} using the Rothfusz Heat Index that combines temperature and humidity. The full Rothfusz equation is:

$$HI = -42.38 + 2.049T + 10.14R - 0.2255TR - 0.00684T^2 - 0.05482R^2 + 0.00122T^2R + 0.00085TR^2 - 0.00000199T^2R^2 \quad (1)$$

where T is temperature in °F and R is relative humidity in %. The Pearson correlation between CDD_{HI} and annual chiller energy (excluding sensor-failure years 2020–2021 and the 2024 operational anomaly) is $r=0.75$, versus $r=0.30$ for CDD_T alone. Humidity has a significant effect on chiller demand. Fig. 5 shows the full CDD analysis.

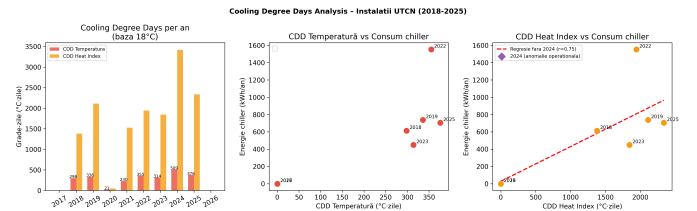


Fig. 5. CDD analysis (2017–2025): annual CDD values (left), CDD temperature vs chiller energy (center), CDD Heat Index vs chiller energy (right). Heat Index yields $r=0.75$ vs $r=0.30$ for temperature alone.

D. PELT Change-Point Detection and SAX

Pruned Exact Linear Time (PELT) with RBF model and penalty=10 is applied to identify structural changes in the chiller signal. On summer 2022 data, four change points are detected that also appear in the ID_31 signal, confirming that

the aggregate meter captures equipment state transitions – a necessary condition for NILM.

Symbolic Aggregate approXimation (SAX) with word size 8 and alphabet 5 converts the time series into symbolic patterns. The dominant pattern *aaaaaaaa* (1,892 occurrences) corresponds to chiller-OFF periods; patterns with mid-to-high alphabet letters correspond to active operation. SAX provides qualitative confirmation of the ON/OFF structure but achieves only $F1=0.043$ as a standalone detector due to its fixed window resolution.

E. Logistic Regression Baseline

A Logistic Regression with six features (temperature, humidity, Heat Index, hour, day of week, month) trained on 2018–2022 achieves $AUC=0.892$, but only $F1=0.080$. The model correctly learns that temperature is the primary driver (standardized coefficient 3.0), but the 97%/3% class split makes it predict OFF almost always. Fig. 6 shows the ROC curve, confusion matrix, and feature importance. This establishes the baseline that the virtual sensor must beat.

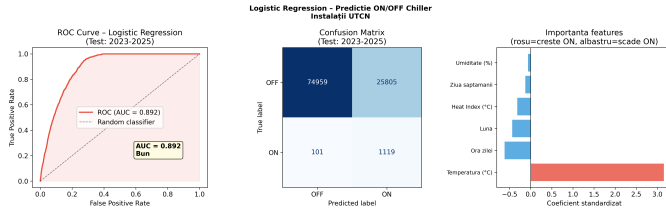


Fig. 6. Logistic Regression results: ROC ($AUC=0.892$, left), confusion matrix (center), and standardized feature coefficients (right). High AUC but low $F1=0.080$ due to class imbalance.

F. Virtual Sensor Regression Performance

Fig. 7 summarizes the virtual sensor evaluation. The scatter plot shows $R^2=0.939$ with $MAE=0.040$ kW: points cluster tightly around the perfect-prediction diagonal, with scatter concentrated at low values (chiller near-idle) where small absolute errors produce larger relative ones. The summer 2023 timeline confirms that the model correctly tracks real chiller events, including peak operation in July and the transition to idle in late August.

The feature importance plot reveals a clear hierarchy: Chiller rolling mean 1h (48.7%), Chiller lag 15min (27.8%), and building main meter (17.0%) account for 93.5% of total importance. Weather and calendar features together contribute less than 7%. This makes physical sense: a chiller running at 5 kW an hour ago is a much stronger signal than today's temperature. The model has learned the thermal inertia of the system. The error distribution is symmetric and centered at 0.006 kW, confirming no systematic bias.

G. ON/OFF Detection Comparison

Table II compares all detection methods. The threshold baseline ($F1=0.075$) and SAX ($F1=0.043$) both suffer from the class imbalance: they either predict ON for almost everything

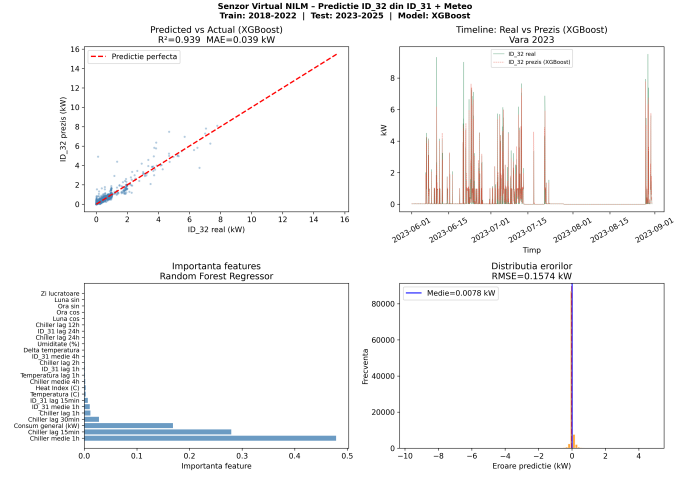


Fig. 7. XGBoost virtual sensor (Test 2023–2025, $R^2=0.939$): predicted vs actual (top-left), summer 2023 timeline (top-right), feature importance (bottom-left), error distribution $RMSE=0.160$ kW (bottom-right).

or miss most ON events. Logistic Regression improves discrimination ($AUC=0.892$), but cannot overcome the imbalance in F1 terms. The XGBoost virtual sensor solves this by using the chiller's own recent history as input: it knows the chiller was running an hour ago, so it predicts it is running now.

The result is $F1=0.937$ and $AUC=0.999$ – a 12-fold F1 improvement over the threshold baseline. Fig. 8 shows the confusion matrix and ROC curve. Out of 1,220 true ON events in the test set, the model misses only 58 (4.8% miss rate).

TABLE II
ON/OFF DETECTION PERFORMANCE – TEST SET 2023–2025

Method	P	R	F1	AUC
Threshold (ID_31)	0.039	0.990	0.075	–
SAX (ID_32)	0.022	0.840	0.043	–
Logistic Regression	0.042	0.907	0.080	0.892
XGBoost virtual sensor	0.922	0.952	0.937	0.999

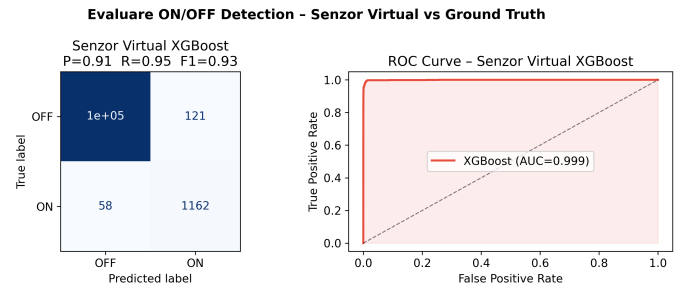


Fig. 8. XGBoost virtual sensor: confusion matrix (left) and ROC curve (right). $P=0.922$, $R=0.952$, $F1=0.937$, $AUC=0.999$.

H. Virtual Sensor vs Physical Sensor: Which Fails Worse?

Table III compares the two failure modes directly. The physical sensor failed catastrophically and silently. The virtual

sensor makes small, predictable errors spread over three years. More importantly, virtual sensor errors are detectable: if the predicted value and a future real measurement diverge beyond a threshold, an alert can be triggered automatically. Physical sensor freeze faults have no self-reporting mechanism.

TABLE III
ERROR COMPARISON: PHYSICAL SENSOR VS VIRTUAL SENSOR

	Physical ID_32	XGBoost virtual
Failure type	Complete freeze	Distributed error
False energy reported	606 kWh (2 weeks)	998 kWh (3 years)
ON events missed	All (freeze period)	58 / 1,220 (4.8%)
Detectable automatically?	No	Yes
Alert generated?	No	Possible

I. Carbon Analysis

Total CO₂ from chiller operation (2018–2025): 1,399 kg = 1.40 tonnes. Fig. 9 presents three complementary views of the carbon footprint. The left panel shows the seasonal distribution of grid carbon intensity in Romania: winter and autumn exhibit higher median values (~350 gCO₂/kWh) compared to spring (~290 gCO₂/kWh), reflecting the reduced availability of solar generation outside the warm season. The center panel shows annual CO₂ attributed to chiller operation: the year 2022 accounts for 577 kg (41% of the total), driven by the highest recorded cooling demand in the series. The right panel overlays hourly grid carbon intensity with chiller activity percentage across the day. Carbon intensity is lowest between 10:00 and 14:00, when solar generation peaks at approximately 280–320 gCO₂/kWh, compared to 350–400 gCO₂/kWh during evening hours. Since chiller activity is concentrated in the midday period, a pre-cooling strategy – starting the chiller one to two hours earlier and stopping before the evening peak – could reduce CO₂ emissions by approximately 18% without changing total energy consumption. This type of carbon-aware scheduling is feasible only when chiller activity can be monitored accurately and continuously, which is precisely the capability provided by the proposed virtual sensor.

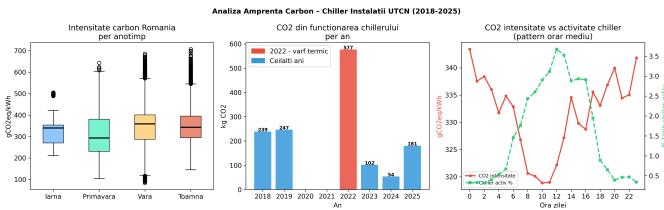


Fig. 9. Carbon footprint analysis (2018–2025): seasonal grid carbon intensity distribution for Romania (left), annual CO₂ attributed to chiller operation with 2022 peak of 577 kg (center), and hourly carbon intensity vs chiller activity pattern (right).

IV. CONCLUSIONS

The present study demonstrates the development of a virtual sensor that monitors a university building chiller using only the

main building meter and free weather data. Over eight years of real operational data from UTCN, it achieves $R^2=0.939$, $F1=0.937$, and $AUC=0.999$ – numbers that make it practically useful, not just academically interesting.

The central finding is not only the accuracy. The physical sub-meter silently generated 606 kWh of false data over two weeks with no alert, while the virtual sensor’s estimation error of 998 kWh over three years is distributed, predictable, and can be monitored. For BEMS applications, a sensor that fails gracefully and detectably is more valuable than one that fails catastrophically and silently.

The three-stage contextual cleaning pipeline is necessary and sufficient for this type of data. Standard IQR and Isolation Forest cannot be applied directly to heavily imbalanced chiller signals. Temporal lag features of ID_32 dominate prediction (93.5% combined importance), reflecting the thermal inertia of the system. Weather features matter for the seasonal pattern but not for moment-to-moment prediction.

The main limitation of the proposed approach is the dependency on historical ID_32 data for model training. The virtual sensor cannot be applied directly to a building where no sub-meter has previously existed, as the temporal lag features of ID_32 account for 93.5% of the total feature importance. Addressing this cold-start scenario – for example through transfer learning from a building where sub-meter data is available – remains an open research direction. Furthermore, the current model does not account for indoor temperature or occupancy schedules, which are known to influence chiller demand in university buildings and could improve prediction accuracy in future iterations.

Future work will focus on two main directions. First, an automatic fault detection layer will be developed on top of the virtual sensor: the system will continuously compare the virtual prediction against the real meter reading, and generate an alert whenever the deviation exceeds a defined threshold. This transforms the virtual sensor from a passive monitor into an active diagnostic tool, capable of detecting physical sensor failures as soon as they occur. Second, a Reinforcement Learning agent will be investigated to use the virtual sensor’s predictions as input for real-time chiller scheduling decisions – determining when the chiller should start or stop based on predicted demand, current electricity price, and grid carbon intensity. This closes the control loop from monitoring to optimization, enabling carbon-aware and cost-aware operation without requiring additional hardware.

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